Senith Jayakody

Research Assistant – Multidisciplinary AI Research Centre, University of Peradeniya

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Research Profile

Electrical and Electronic Engineer and Research Assistant working at the intersection of machine learning, signal processing, computational imaging, sensing hardware, and embedded systems. Research experience includes biomedical time-series analysis, multispectral imaging for environmental and agricultural monitoring, and diffusion-based generative modelling. Experienced across the end-to-end research pipeline: experimental design, data acquisition, feature engineering, model development and validation, hardware integration, technical writing, and publication.

Research Interests

Machine learning for sensing systems; biomedical signal processing and physiological time-series analysis; computational and multispectral imaging; robust and interpretable learning; generative and diffusion models; embedded and edge AI; environmental and agricultural monitoring.

Education

University of Peradeniya *2019 – 2025*
B.Sc. Eng. (Hons.) in Electrical and Electronic Engineering Peradeniya, Sri Lanka

- **First Class Honours**; GPA: **3.85/4.00**.
- Degree programme conducted and assessed in English; programme accredited by the Institution of Engineers, Sri Lanka under the Washington Accord framework.
- Academic Transcript

Trinity College Kandy *2019*
G.C.E. Advanced Level Examination – Physical Science Stream Kandy, Sri Lanka

- Three Distinction (A) passes in Combined Mathematics, Physics, and Chemistry; placed among the top 40 candidates in Kandy District and top 550 nationwide.
- Advanced Level Result Sheet

Trinity College Kandy *2016*
G.C.E. Ordinary Level Examination Kandy, Sri Lanka

- Nine Distinction (A) passes. Ordinary Level Result Sheet

Research and Professional Experience

Multidisciplinary AI Research Centre (MARC), University of Peradeniya *2025 – Present*
Research Assistant Peradeniya, Sri Lanka

- Lead research on preterm-birth prediction using electrohysterography and tocographic signals, focusing on spectrotemporal features, model robustness, interpretability, and reproducible evaluation.
- Design a dedicated biomedical data-acquisition system to collect term and preterm pregnancy recordings and address dataset scarcity.
- Supervise junior researchers in multispectral-imaging experiments, algorithm development, and data analysis.
- Contribute to calibration, illumination, reliability, and embedded integration of a field-deployable multispectral-imaging platform.

Multidisciplinary AI Research Centre (MARC), University of Peradeniya *2023 – 2025*
Undergraduate Researcher Peradeniya, Sri Lanka

- Developed reflectance multispectral-imaging models for soil-moisture estimation and evaluated robustness across multiple soil conditions.
- Conducted soil-composition estimation, USDA texture classification, and *Microcystis* cell-count and toxicity-estimation studies.
- Improved the imaging platform through illumination redesign, circuit updates, camera control, and Raspberry Pi-based integration.

Swartup Pvt. Ltd. *Jul 2024 – Oct 2024*
Research and Development Intern Sri Lanka

- Migrated embedded systems from ESP32/FreeRTOS to NRF5340/Zephyr and designed modular platforms with LoRa and GSM capability.

Sri Lanka Telecom Digital Lab *Jul 2023 – Oct 2023*
Research and Development Intern Sri Lanka

- Developed IoT-based environmental-monitoring systems using ESP32, MQTT, Firebase, and integrated sensors.

Publications

Preprints and manuscripts

[J1] **Jayakody, S.**, Jayasooriya, K., Liyanage, S., Godaliyadda, R., Ekanayake, P., Nawinne, I., and Rathnayake, C., "Spectrotemporal feature extraction in ehg signals and tocograms for enhanced preterm birth prediction," Manuscript submitted to IEEE Access; under review, 2025. arXiv: 2509.07432

[P1] Ranasinghe, G. A. S. L., **Jayakody, J. A. S. T.**, De Silva, M. C. L., Thilakarathne, G., Godaliyadda, G. M. R. I., Herath, H. M. V. R., Ekanayake, M. P. B., and Navaratnarajah, S. K., *Reflectance multispectral imaging for soil composition estimation and usda texture classification*, 2026. arXiv: 2602.22829 [cs.CV]

Peer-reviewed conference proceedings

[C1] Perera, S., Ranasinghe, S., **Jayakody, S.**, Epakanda, B., Godaliyadda, R., and Ekanayake, M. P., "Feature conditioned diffusion for audio generation," in *2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, vol. 19, 2026, pp. 162–167. DOI: 10.1109/ICIIS69028.2026.11450741

[C2] Thilakarathne, G., Perera, S., Ranasinghe, S., **Jayakody, S.**, Godaliyadda, R., Herath, V., Ekanayake, M. P., and Navaratnarajah, S., "Multispectral imaging and machine learning for soil classification through moisture-induced spectral dynamics," in *2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, vol. 19, 2026, pp. 569–574. DOI: 10.1109/ICIIS69028.2026.11450634

[C3] Ranasinghe, S., **Jayakody, S.**, Silva, M. D., Herath, V., Godaliyadda, R., Ekanayake, M. P., Navaratnarajah, S., Kizel, F., and Thilakarathne, G., "Deep learning for soil moisture content estimation via reflectance multispectral imaging," in *2024 International Conference on Advances in Technology and Computing (ICATC)*, 2024, pp. 1–6. DOI: 10.1109/ICATC64549.2024.11025290

Selected Research Projects

Preterm-Birth Prediction Using EHG and TOCO Signals *2025 – Present*
Biomedical signal processing, feature engineering, machine learning, data acquisition

- Developed spectrotemporal feature-extraction and classification pipelines for term/preterm prediction, emphasizing robustness, clinical interpretability, and reproducible validation.
- Designed a dedicated acquisition platform to support collection of clinically relevant pregnancy recordings.
- Preprint available through the publication entry above.

Multispectral Imaging for Agricultural and Environmental Monitoring *2023 – Present*
Computational imaging, regression, classification, calibration, embedded systems

- Developed models for soil-moisture estimation, soil-composition prediction, USDA texture classification, harmful-algal-bloom assessment, and food-adulteration studies.
- Integrated illumination, camera control, calibration, and embedded processing into a reflectance/transmittance imaging platform.
- Contributed to experimental design, dataset development, validation, and publication preparation across multiple applications.

Feature-Conditioned Diffusion for Audio Generation *2025*
Diffusion models, audio signal processing, temporal representation learning

- Investigated signal-processing-derived temporal conditioning for DiffWave-based generation to improve waveform stability and temporal coherence.

Selected Engineering Prototypes *2022 – 2024*
Optical sensing, sensor fusion, embedded systems, real-time firmware

- Developed a non-invasive optical blood-glucose sensing prototype, a Kalman-filter-based terrain-mapping system (Survey Buddy), and an assembly-language cyclist-safety system (GloSmart).

Honors and Awards

Second Runner-Up, Manamperi Award – Best Undergraduate Project in Sri Lanka; multispectral-imaging final-year project *2025*
Silver Award, Engineering Category – Sahasak Nimavum Inventions and Innovations Competition *2024*
Top 10 Finalist – Predicta v1.0, IEEE Student Branch, University of Peradeniya *2024*
Finalist – IEEE Electronic Design Competition, IEEE Sri Lanka Section *2023*
Second Place – ACES Hackathon, University of Peradeniya *2023*
Fourth Place – IESL Robogames, University Category *2022*

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Assembly
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, feature engineering, model evaluation and validation
- **Signal and Image Processing:** Time-frequency and spectral analysis, MFCC, statistical modelling, OpenCV
- **Embedded and Sensing Systems:** Raspberry Pi, ESP32, NRF5340, firmware development, sensor integration, circuit design, calibration
- **Tools:** Linux, Git, MQTT, Firebase, Altium Designer, EasyEDA, LaTeX

Academic Service and Outreach

Workshop Facilitator – Conducted AI and embedded-systems sessions for pre-engineering students at the Diamond Jubilee AI Workshop *2025*

Research Demonstrator and Speaker – Presented real-world AI and multispectral-imaging applications at the Trinity College AI Symposium and DreamEx *2025*

Logistics Team Member – Supported robotics competitions and technical workshops, IEEE Robotics and Automation Society Student Chapter *2021 – 2022*

Volunteer – Supported technical sessions and student-engagement activities, IET on Campus *2022 – 2023*

Selected Professional Training

- **2024:** Unsupervised Learning, Recommenders, and Reinforcement Learning (Stanford University/DeepLearning.AI, Coursera); Algorithmic Toolbox (University of California San Diego, Coursera); Introduction to Computer Vision and Generative AI for Beginners (Great Learning).
- **2023:** Advanced Learning Algorithms; Supervised Machine Learning: Regression and Classification (Stanford University/DeepLearning.AI, Coursera).
- **2021:** Programming for Everybody (University of Michigan, Coursera); Engineering Drawing and 3D Modelling using AutoCAD (TecView Institute).

Professional Membership

Student Member, Institution of Engineers, Sri Lanka (IESL), Membership No. S-31245.

Academic References

1. **Prof. G. M. R. I. Godaliyadda**
   Ph.D., National University of Singapore; B.Sc. Eng. (Hons.), University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering, University of Peradeniya
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2. **Prof. H. M. V. R. Herath**
   Dr.-Ing., University of Paderborn, Germany; M.S., University of Miami, USA; B.Sc. Eng., University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering, University of Peradeniya
   vijitha@ee.pdn.ac.lk | Office: +94 81 239 3436 | Mobile: +94 71 673 2688

3. **Prof. M. P. B. Ekanayake**
   Ph.D. in Applied Mathematics, Texas Tech University, USA; B.Sc. Eng. (Hons.), University of Peradeniya
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